

Lesson: 01

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Mohsin Ibna Hossain
AIUB, Physics 1 Note,
Mid Term:—

Day : 

Time :

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Ch 04
Motion in Two and
Three Dimensions.

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Rest and Motion :-

1. Motion is Combined Property of an Object under study and the observer.
2. Everything in the world is at rest and motion.
3. There is no meaning of rest and motion without the viewer.

Relative Rest And Motion :-

⇒ Rest and Motion are relative terms.

Ex:- A Person who is sitting in a moving bus is at rest because he is not changing his position with respect to other person in the bus. But to an observer out side the bus, the persons and the all thing inside the bus are in motion.

Force :-

⇒ Force is a physical quantity that describes the interaction between objects that result change in a change in motion.

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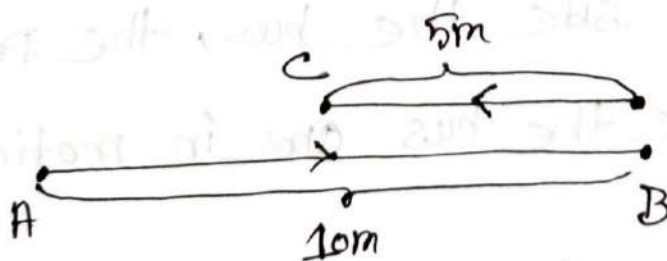
Force Can do three thing:-

1. Rest body start to move.
2. On a Moving body can be Stopped.
3. Change the shape of any things.

Q What is Reference frame?

⇒ The Observer relative to which any motion on rest is determined is called reference frame.
It's can may be a Person, an object on a place.

#Distance & Displacement:-



⇒ If you moves A Point to B Point. Then you turns round and moves C Point.

Then your total distance is $= 10 + 5 = 15\text{m}$

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But your Displacement is $= 10 - 5 = 5\text{m}$.

Speed And Velocity:—

We know,

$$S = \frac{d}{t}, \quad S = Vt$$

Where,

$S = \text{Speed}$

$d = \text{Distance}$

$t = \text{time}$

Where,

$S = \text{Displacement}$

$V = \text{Velocity}$

$t = \text{time}$

So, $S = \frac{\text{Distance}}{\text{time}}$

So, $V = \frac{\text{Distance in a Particular direction}}{\text{time}}$

Acceleration And Retardation:—

$\Rightarrow \text{Acceleration} / a = \frac{\text{total increase in Velocity}}{\text{total time}}$

It means

$$a = \frac{\Delta V}{\Delta t} / \frac{v_2 - v_1}{t_2 - t_1} / \frac{v_f - v_i}{t_f - t_i}$$

avg $\lim_{\Delta t \rightarrow 0} \frac{\Delta V}{\Delta t} = \frac{dV}{dt}$

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$$\text{Retardation} = \frac{\text{Total decrease in Velocity}}{\text{Total time.}}$$

Basic: 01

Let, Starting time = t_1

Ending time = t_2

Initial Velocity = $\vec{V}_1$

final velocity = $\vec{V}_2$

$$\therefore \vec{a} = \frac{\vec{V}_2 - \vec{V}_1}{t_2 - t_1}$$

$$\vec{a} = \frac{d\vec{V}}{dt}$$

Basic: 02

Let, Starting time = t_1

Ending time = t_2

Initial position = $\vec{S}_1$

Final position = $\vec{S}_2$

$$\therefore \text{Velocity } / \vec{V} = \frac{\vec{S}_2 - \vec{S}_1}{t_2 - t_1}$$

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Variables: _____

1. Independent Variable.
2. Dependent Variable.

Ex:- $Y = mx$

x = Independent Variable.

Y = Dependent Variable.

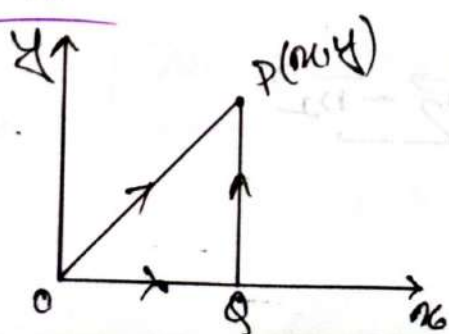
x	1	2	3	4
Y	1	2	3	4

Position: _____

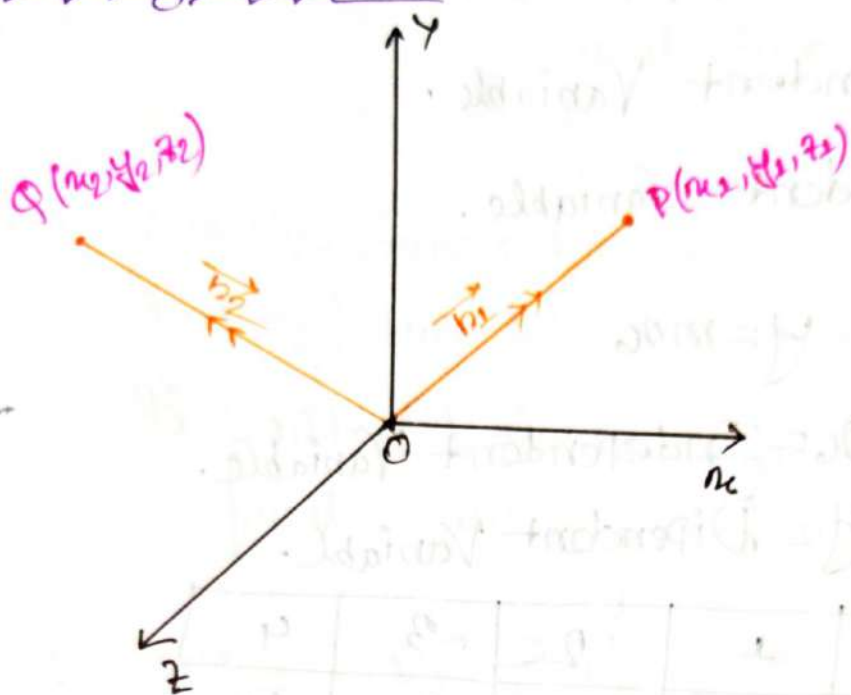
One Dimension: _____



Two Dimensions: _____



$$\vec{OP} = x\hat{i} + y\hat{j}$$

Three Dimensions

$$\vec{OP} = \vec{r}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

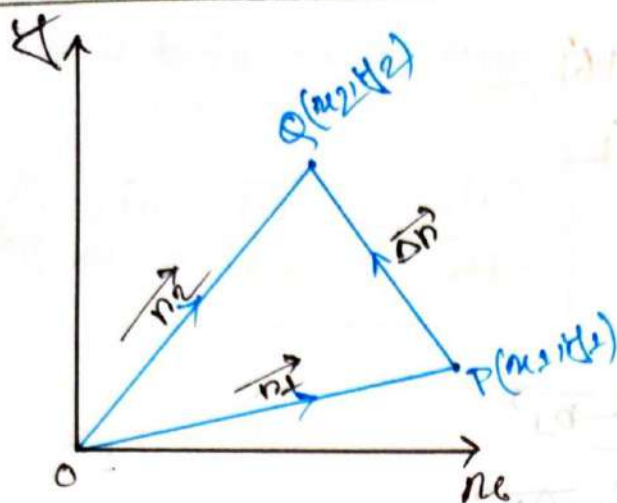
$$\vec{OQ} = \vec{r}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

Position Vector and Displacement Vector

During a time interval Δt the particle moves from P_1 , where its Position Vector is $\vec{r}_1$ to P_2 , where its Position Vector is $\vec{r}_2$. The change in position during this interval is

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$





$$\vec{OP} = \vec{r}_1 = x_1 \hat{i} + y_1 \hat{j}$$

$$\vec{OQ} = \vec{r}_2 = x_2 \hat{i} + y_2 \hat{j}$$

We know,

$$\vec{OP} + \vec{PQ} = \vec{OQ}$$

$$\therefore \vec{PQ} = \Delta \vec{r} = \vec{OQ} - \vec{OP} = \vec{r}_2 - \vec{r}_1$$

$$= (x_2 \hat{i} + y_2 \hat{j}) - (x_1 \hat{i} + y_1 \hat{j})$$

$$= (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j}$$

= Displacement

in three Dimensions —

$$\Delta \vec{r} = \text{Displacement} = \vec{r}_2 - \vec{r}_1$$

$$= (x_2 \hat{i} + y_2 \hat{j} + z_2 \hat{k}) - (x_1 \hat{i} + y_1 \hat{j} + z_1 \hat{k})$$

$$\Delta \vec{r} = \hat{i}(x_2 - x_1) + \hat{j}(y_2 - y_1) + \hat{k}(z_2 - z_1)$$

Q:- $\Delta \vec{r} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

$\vec{r}_2 = 3\hat{j} - 4\hat{k}$

$\vec{r}_1 = ?$

$\Rightarrow$ We know,

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\therefore \vec{r}_1 = \vec{r}_2 - \Delta \vec{r}$$

$$= (3\hat{j} - 4\hat{k}) - (2\hat{i} - 3\hat{j} + 6\hat{k})$$

$$= -2\hat{i} + 3\hat{j} + 3\hat{j} - 10\hat{k}$$

$$= -2\hat{i} + 6\hat{j} - 10\hat{k}$$

Ans

Average Velocity And

Instantaneous Velocity:-

$$\vec{V}_{avg} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta \vec{r}}{\Delta t}$$

$$\vec{r} = (x\hat{i} + y\hat{j} + z\hat{k})$$

Instantaneous Velocity:

$$\vec{V} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{h}}{\Delta t} = \frac{d\vec{h}}{dt}$$

Q. $\vec{h}(t) = x\hat{i} + y\hat{j} + z\hat{k}$

We know, $\vec{V} = \frac{d\vec{h}}{dt}$

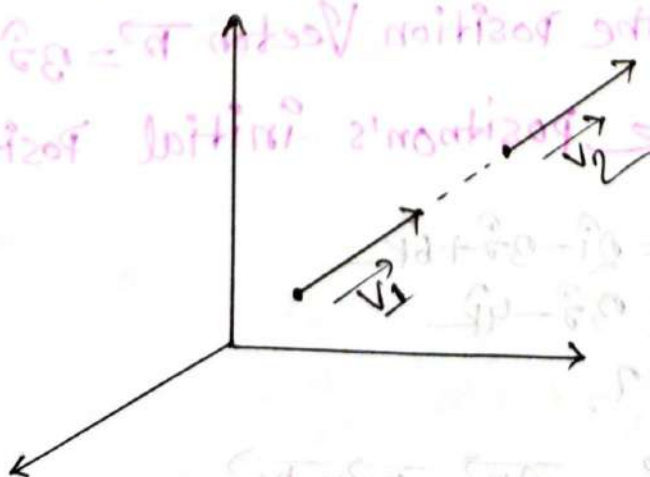
$$\therefore \vec{V} = \frac{d}{dt} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$= \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

$$= v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$$

Vector Magnitudes

$$* |\vec{V}| = V = \sqrt{v_x^2 + v_y^2 + v_z^2}$$



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$$\vec{V}_1 = v_{x1}\hat{i} + v_{y1}\hat{j} + v_{z1}\hat{k}$$

$$\vec{V}_2 = v_{x2}\hat{i} + v_{y2}\hat{j} + v_{z2}\hat{k}$$

$$\therefore \Delta\vec{V} = \vec{V}_2 - \vec{V}_1$$

$$= (v_{x2} - v_{x1})\hat{i} + (v_{y2} - v_{y1})\hat{j} + (v_{z2} - v_{z1})\hat{k}$$

$$\therefore |\vec{V}_1| = v_1 = \sqrt{(v_{x1})^2 + (v_{y1})^2 + (v_{z1})^2}$$

$$|\vec{V}_2| = v_2 = \sqrt{(v_{x2})^2 + (v_{y2})^2 + (v_{z2})^2}$$

Vector Direction :-

$$\tan\alpha = \frac{v_y}{v_x}$$

$$\text{or } \alpha = \tan^{-1}\left(\frac{v_y}{v_x}\right)$$

Q:- A position Undergoes a Displacement $\Delta\vec{r} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ ending with the position Vector $\vec{r}_2 = 3\hat{j} - 4\hat{k}$, in meters. What was the position's initial position Vector $\vec{r}_1$.

$$\Rightarrow \text{Given, } \Delta\vec{r} = 2\hat{i} - 3\hat{j} + 6\hat{k}$$

$$\vec{r}_2 = 3\hat{j} - 4\hat{k}$$

$$\vec{r}_1 = ?$$

We know,

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\therefore \vec{r}_1 = \vec{r}_2 - \Delta\vec{r}$$

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$$\therefore \vec{r}_1 = (3\hat{i} - 4\hat{k}) - (2\hat{i} - 3\hat{j} + 6\hat{k})$$

$$= -2\hat{i} + 6\hat{j} - 10\hat{k}$$

Ans:

Q.1 A Particle moves so that its Position as a function of time is $\vec{r} = \hat{i} + 4t^2\hat{j} + t\hat{k}$. Write expressions for.

(a) its Velocity and

(b) its acceleration as functions of time.

$\Rightarrow$ is given, $\vec{r} = \hat{i} + 4t^2\hat{j} + t\hat{k}$

We know $\vec{v} = \frac{d\vec{r}}{dt}$

$$\therefore \vec{v} = \frac{d}{dt} (\hat{i} + 4t^2\hat{j} + t\hat{k})$$

$$= 0 + 2 \times 4t\hat{j} + \hat{k}$$

$$= 8t\hat{j} + \hat{k}$$

(b)

We know,

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$= \frac{d}{dt} (8t\hat{j} + \hat{k})$$

$$= 8\hat{j} + 0$$

$$|\vec{a}| = 8 \text{ m/s}^2$$

Ans

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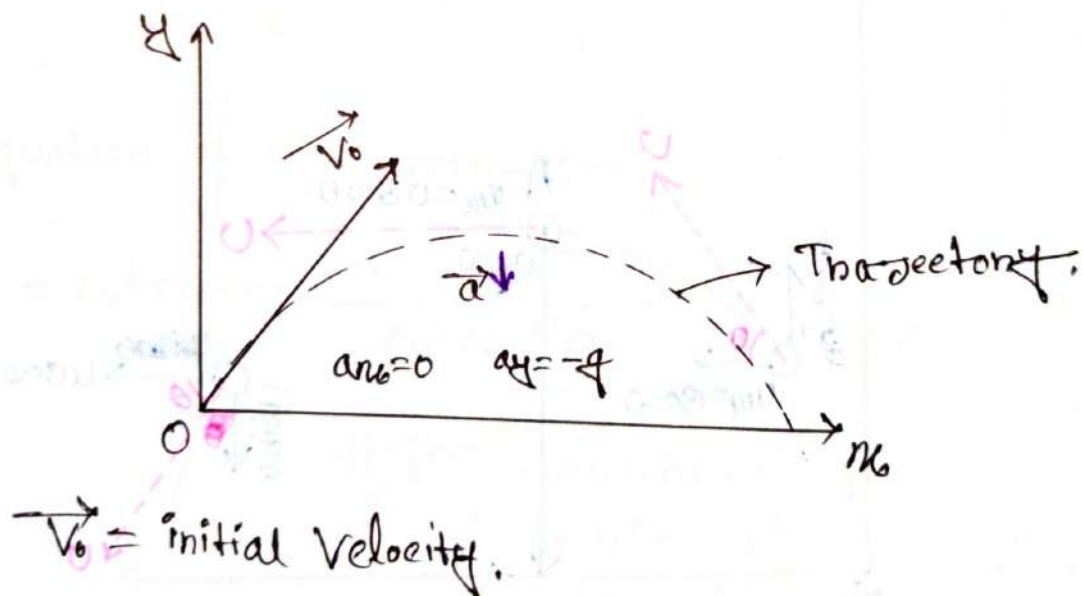
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#Projectile Motion

#Projectile motion

⇒ Projectile motion is the motion of an object that is projected into the air and then is subject to only the force of gravity. It is a two dimensional motion, where an object moves in a Parabolic Path under the influence of the Earth's gravity. The object has both horizontal and Vertical Velocity Components, and the path is determined by the initial Velocity, angle, and the gravitational acceleration.



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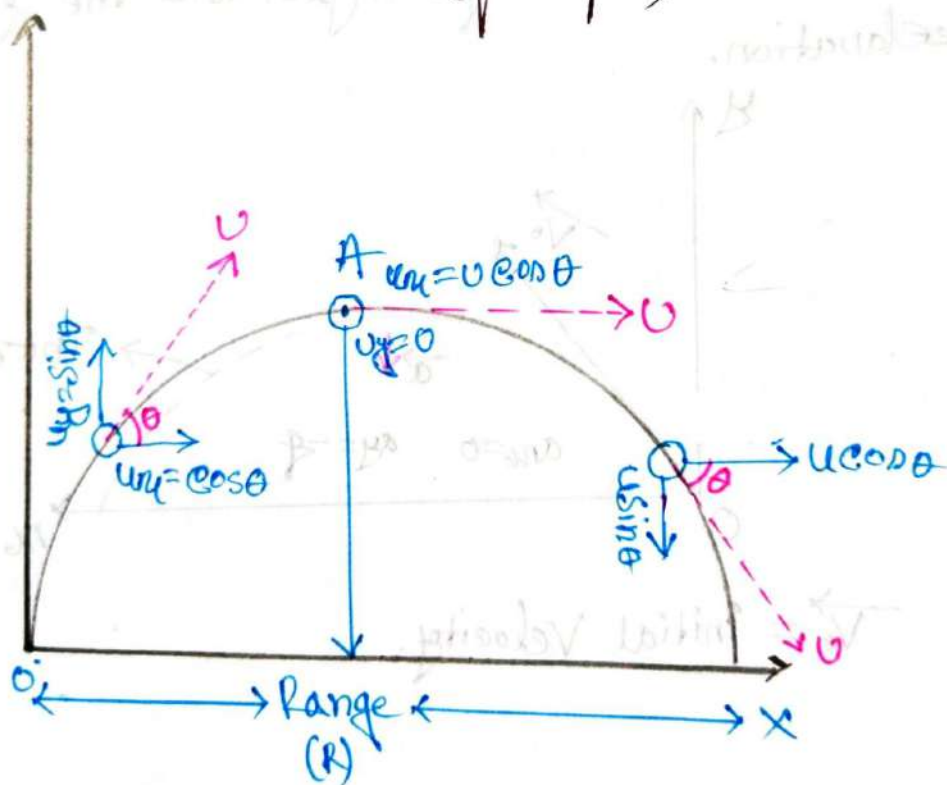
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Conditions of Motion of a Projectile:—

1. Should be thrown with an acute angle from ground.
2. Motion will be two dimension.
3. Same Plane.
4. Uniform Acceleration (free fall).



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* Total time of flight (t)

$$t = \frac{2u \sin \theta}{g}$$

* Maximum Height ($H_{\max}$)

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

* Horizontal range (R)

$$R = \frac{u^2 \sin 2\theta}{g}$$

* Equation of Trajectory:—

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

* The Horizontal Motion:—(R)

⇒ At any time t , the Projectile's horizontal displacement $x - x_0$ from an initial position x_0 is given by.

$$x - x_0 = v_{0x}t + \frac{1}{2}a_x t^2$$

Where $a_x = 0$

Then,

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$$u_x - u_{x0} = v_{0x} t$$

$$R = (v_0 \cos \theta_0) t$$

$$\text{Range (R)} = (v_0 \cos \theta_0) t$$

at any time t , the horizontal velocity component

$$v_{hx} = v_{ox} = v_0 \cos \theta_0$$

100% correct

The Vertical Motion:

⇒ At any time t , the Projectile's Vertical displacement $y - y_0$ from an initial position y_0 is given by

$$\text{So } y - y_0 = v_{0y} t - \frac{1}{2} g t^2$$

$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2 \quad [\text{where } v_{0y} = v_0 \sin \theta_0]$$

At any time t , the Projectile's Vertical velocity component.

$$v_y = v_0 \sin \theta_0 - g t$$

$$v = u + at$$

$$v_y^2 = (v_0 \sin \theta_0)^2 - 2g(y - y_0)$$

$$v^2 = u^2 \pm 2as$$

100% correct

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Show that the path of a Projectile is a Parabola:

⇒ We know,

$$x - x_0 = (v_0 \cos \theta_0) t$$

$$\therefore t = \frac{x - x_0}{v_0 \cos \theta_0} \quad \text{--- (i)}$$

Again:-

$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$$

$$= (v_0 \sin \theta_0) \times \frac{x - x_0}{v_0 \cos \theta_0} - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos \theta_0} \right)^2$$

$$= \tan \theta_0 \times x - \frac{1}{2} g \left(\frac{x}{v_0 \cos \theta_0} \right)^2 \quad [\text{when } x_0 = 0]$$

$$= \tan \theta_0 \times x - \frac{g x^2}{2 v_0^2 \cos^2 \theta_0}$$

$$= \tan \theta_0 \times x - \frac{1}{2} \times \frac{g}{v_0^2 \cos^2 \theta_0} \times x^2$$

$$= \tan \theta_0 \times x + \left(- \frac{g}{2 v_0^2 \cos^2 \theta_0} \right) x^2 \quad \text{--- (ii)}$$

Where θ_0 , g and v_0 are constants

Equation (ii) is of the form $y = ax + bx^2$, where a and b constants.

— This is the equations of Parabola, So the Path is Parabolic.

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Equations for the horizontal range and the maximum horizontal range of a Projectile: —————

⇒ when $y - y_0 = 0$

That is $x_0 - x_0 = R$

we know,

$$R = (V_0 \cos \theta_0) t \quad \text{————— (1)}$$

and $y - y_0 = (V_0 \sin \theta_0) t - \frac{1}{2} g t^2$

on, $0 = (V_0 \sin \theta_0) t - \frac{1}{2} g t^2$

So, $t = \frac{2V_0 \sin \theta_0}{g} \quad \text{————— (2)}$

now,

$$R = (V_0 \cos \theta_0) \times \frac{2V_0 \sin \theta_0}{g}$$

$$R = \frac{V_0^2 \sin 2\theta_0}{g} \quad \left[2 \sin \theta_0 \times \cos \theta_0 = \sin 2\theta_0 \right]$$

$$\therefore R = \frac{V_0^2 \sin 2\theta_0}{g}$$

when the value of R is max in this equation.

$$\sin 2\theta = 1$$

on, $2\theta = \sin^{-1}(1)$

on, $\theta = \frac{90^\circ}{2}$
 $= 45^\circ$

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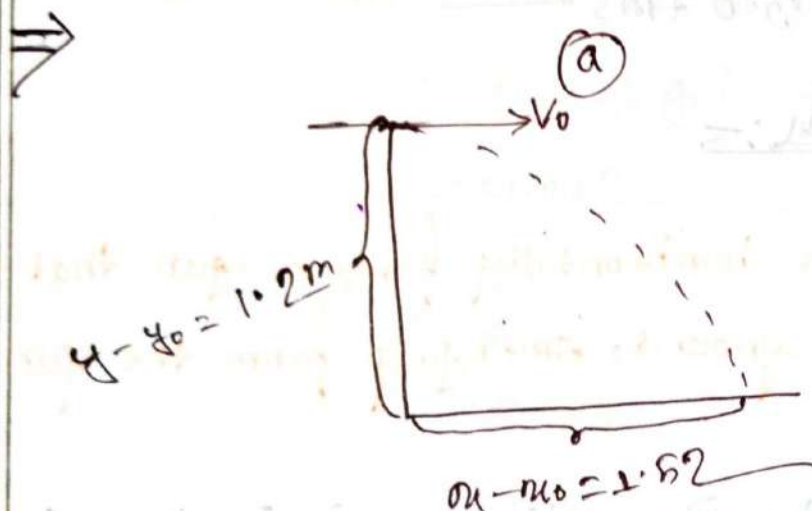
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Q2 - A small ball rolls horizontally of the edge of a table top that is 1.2m height. It strikes ^{the floor} at a point 1.52m horizontally from the table edge.

- (a) How long is the ball in the air?
 (b) What is the speed at the instant it leaves the table?



is given, $y - y_0 = 1.2\text{m}$

$x - x_0 = 1.52\text{m}$

We know,

$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$$

$$-1.20 = (v_0 \sin \theta_0) t - \frac{1}{2} (9.8) t^2$$

$$-1.20 = 0 - 4.9 t^2$$

$$\therefore t = \sqrt{\frac{1.20}{4.9}} = 0.495\text{s}$$

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(b)

We know,

$$R_0 - R_0 = V_0 \cos \theta_0 \times t$$

$$1.52 = V_0 \cos 0^\circ \times (0.405)$$

$$2: V_0 \times 1 \times (0.405)$$

$$\therefore V_0 = \frac{1.52}{0.405}$$

$$= 3.07 \text{ ms}^{-1}$$

Ans :-

Q. A projectile is fired horizontally from a gun that is 45.0 m above flat ground, emerging from the gun with a speed of 250 ms⁻¹.

- (a) How long does the projectile remain in the air.
- (b) At what horizontal distance from the firing point does it strike the ground.
- (c) What is the magnitude of the vertical component of its velocity as it strikes the ground.

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is given,

$$y - y_0 = 45.0 \text{ m}$$

$$v_0 = 250 \text{ m/s}$$

we know,

$$y - y_0 = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2$$

$$\Rightarrow -45.0 = 250(v_0 \sin \theta_0)t - \frac{1}{2}gt^2$$

$$\begin{aligned} \Rightarrow -45 &= (v_0 \sin 0^\circ)t - \frac{1}{2}gt^2 \\ &= 0 - \frac{1}{2}gt^2 \\ &= -4.9t^2 \end{aligned}$$

$$\begin{aligned} \therefore t &= \sqrt{\frac{45}{4.9}} \\ &= 3.0304 \text{ s} \end{aligned}$$

(b)

we know,

$$\begin{aligned} x - x_0 &= (v_0 \cos \theta_0)t \\ &= (250 \times \cos 0^\circ)t \times 3.0304 \\ &= 757.6 \text{ m} \end{aligned}$$

(c)

we know,

$$\begin{aligned} v_y &= v_0 \sin \theta_0 - gt \\ &= 250 \times \sin 0^\circ - 9.8 \times 3.0304 \\ &= -29.6972 \text{ m/s} \end{aligned}$$

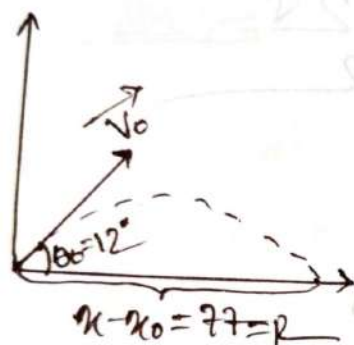
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The magnitude of V_y is 20.69 m/s

Q:- The current world record motorcycle jump is 77.0 m set by Jason Renie. Assume that he left the take off ramp at 12.0° to the horizontal and that the take off landing heights are the same. Neglecting air drag, determine his take off speed.



we know,

$$R = \frac{V_0^2 \sin 2\theta_0}{g}$$

$$\text{on, } Rg = \frac{V_0^2 \sin 2\theta_0}{1}$$

$$\text{on, } V_0 = \sqrt{\frac{Rg}{\sin 2\theta_0}}$$

$$= \sqrt{\frac{77 \times 9.8}{\sin 2 \times 12}}$$

$$= 43.07 \text{ m/s}$$

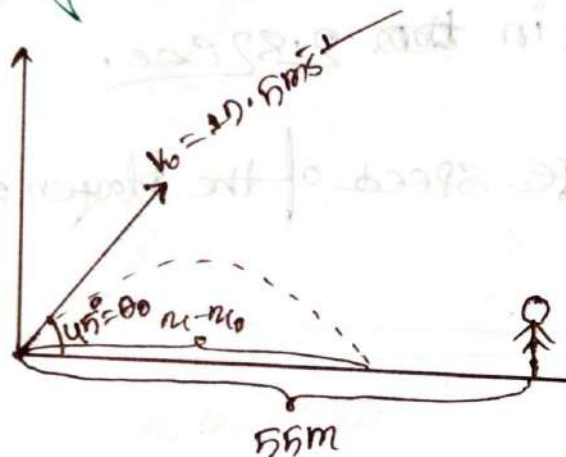
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Q:- A Soccer ball is kicked from the ground with an initial speed of 10.5 m/s at an upward Angle 45° . A Player 55 m Away in the direction of the kick starts running to meet the ball at that instant. What must be his Average speed he is to meet just before ball hits the ground.

⇒



is given,

$$V_0 = 10.5 \text{ m/s}$$

$$\theta_0 = 45^\circ$$

we know,

$$y - y_0 = (V_0 \sin \theta_0)t - \frac{1}{2}gt^2$$

$$\text{on, } 0 = V_0 \sin \theta_0 t - \frac{1}{2}gt^2$$

$$\text{on, } 4.9t^2 = V_0 \sin \theta_0 t$$

$$\text{on, } 4.9t = V_0 \sin \theta_0$$

$$\text{on } t = \frac{10.5 \times \sin 45^\circ}{4.9}$$

$$= 2.82 \text{ Sec}$$

that case,

$$y - y_0 = 0$$

∴ we use the following formula the time of flight of the ball.

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now we need the value of Range

we know,

$$R = (V_0 \cos \theta_0) \times t$$

$$u - u_0 = (10.5 \times \cos 45) \times 2.82 \\ = 38.88 \text{ m}$$

So the player need to overcome $= (55 - 38.88 \text{ m}) = 16.11$ m this range in 2.82 sec.

So, Average speed of the player = $\frac{\text{Distance}}{\text{time.}}$


$$= \frac{16.11}{2.82}$$

$$= 5.712 \text{ m/s}$$

Ans:-

Q:- You throw a ball toward a wall at speed 25 m/s at the Angle 40.0° Above the horizontal. The wall is distance $d = 22.0 \text{ m}$ from the release point of the ball.

a) How far above the release point does the ball hit the wall.

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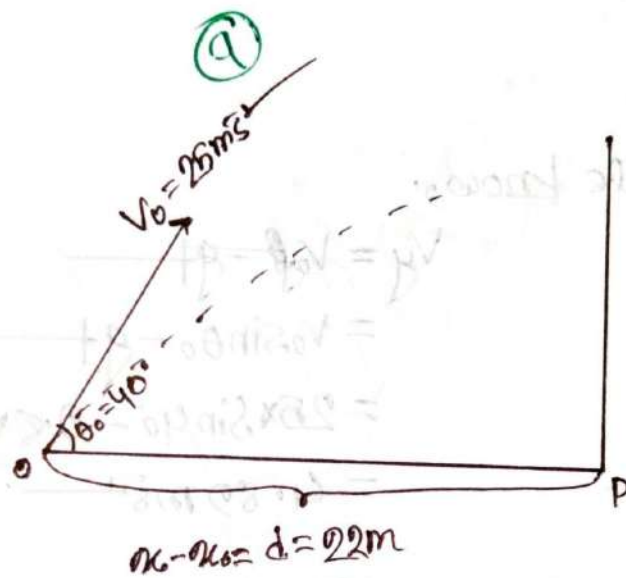
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(b) What are the horizontal and vertical components of its velocity as it hits the wall.

(c) When it hits, has it passed the height point on its trajectory?



We know,

$$x_0 - x_0 = (v_0 \cos \theta_0) t$$

$$\therefore t = \frac{x_0 - x_0}{v_0 \cos \theta_0}$$

$$= \frac{22\text{m}}{25 \times \cos 40^\circ}$$

$$= 1.14 \text{ sec}$$

Again, we know,

$$y - y_0 = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$$

$$= (25 \sin 40^\circ) \times 1.14 - \frac{1}{2} \times 9.8 \times (1.14)^2$$

$$= 12.14 \text{ m}$$

Ans

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(b)

Horizontal

We know,

$$\begin{aligned} V_{x0} &= V_{0x} = V_0 \cos \theta_0 \\ &= 25 \times \cos 40^\circ \\ &= 19.15 \text{ m/s} \end{aligned}$$

Vertical

We know,

$$\begin{aligned} V_y &= V_{0y} - gt \\ &= V_0 \sin \theta_0 - gt \\ &= 25 \times \sin 40^\circ - 9.8 \times (1.14) \\ &= 4.89 \text{ m/s} \end{aligned}$$

(c)

The highest point is reached when the component of Velocity is zero, which occurs before the ball reaches the wall. Since V_y is positive, that is $V_y > 0$, that means the ball did not reach to the highest point on hitting the wall.

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Force and Motion:-1

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Force:

⇒ 1. A force is a push or a pull.

2. A force is an interaction between two objects.

3. A force is a vector quantity, with magnitude and direction.

Units of force:

1. SI Unit: Newton (N);

2. CGS Unit: dyne;

3. British Unit: Pound (lb)

⇒ if two or more forces act on a body, we find the net force by adding them as vectors.



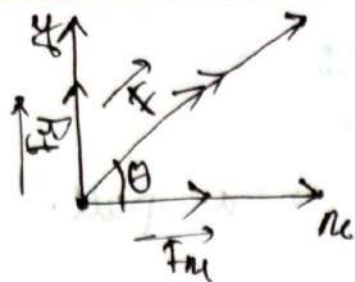
$$\vec{F} = \vec{F}_x + \vec{F}_y$$

→ The force, which acts at an angle from the x-axis, may be replaced by the rectangular component vectors $\vec{F}_x$ and $\vec{F}_y$. Here, $F_x = F \cos \theta$, $F_y = F \sin \theta$.

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$$\therefore \vec{F}_{\text{net}} = F_x \hat{i} + F_y \hat{j}$$

$$= (F \cos \theta) \hat{i} + (F \sin \theta) \hat{j}$$

Some Particular forces:-

* The gravitational force:-

⇒ Gravity is a force that pulls objects towards each other. The bigger the objects and the closer they are, the stronger the force of gravity between them.

the magnitude of $\vec{F}_g$ is

$$F_g = mg$$

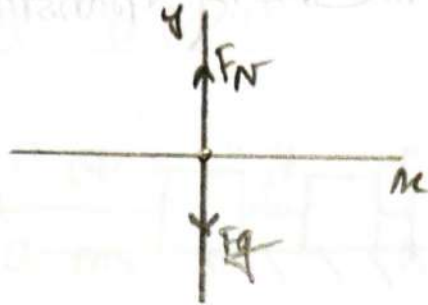
where,

m = body's mass

g = free-fall acceleration.

Normal force:

⇒ A Normal force F_N is the force on a body from a surface against which the body presses. The normal is always perpendicular to the surface.

Weight:

⇒ The weight W of a body is equal to the magnitude F_g of the gravitational force on the body.

That is $W = F_g = mg$

Frictional force:

⇒ Frictional force is a force that opposes motion between two objects in contact and results from the interaction of these surfaces.



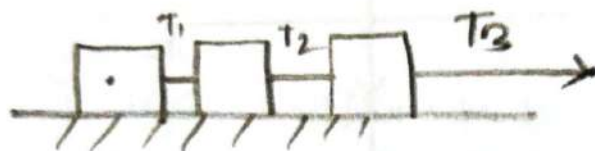
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Tension:—

⇒ Tension is a force exerted along the length of a rope, cable, or other object under tension, equal to the magnitude of the force being transmitted along the objects.



Newtonian Mechanics:—

Newton's First law:—

⇒ If no net force acts on a body, the body's velocity can't change; that is cannot accelerate.

Newton's Second law:—

⇒ The net force on a body is equal to the product of the body's mass (m) and its acceleration (a)

in Vector Equation form,

$$\vec{F}_{\text{net}} = m\vec{a}$$



$$F_{\text{net}} = m\vec{a} \rightarrow \text{if } F_1 > F_2$$

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Basic:

let, Final momentum = mv

Initial momentum = mu

So, $F \Delta t = \frac{mv - mu}{+}$

on, $F \Delta t = \frac{m(v - u)}{+}$

on, $F \Delta t = m \times \frac{v - u}{+}$

on, $F \Delta t = ma$

$\therefore F = ma$

Explain.

Q:

Newton's Second law using Newton's First law.

$\Rightarrow$ we know,

$$F = \frac{mv - mu}{+}$$

for 1st law $F_{\text{net}} = 0$

So, $\frac{mv - mu}{+} = 0$

on, $mv - mu = 0$

on, $mv = mu$

on, $v = u$

that is there is no change on velocity.

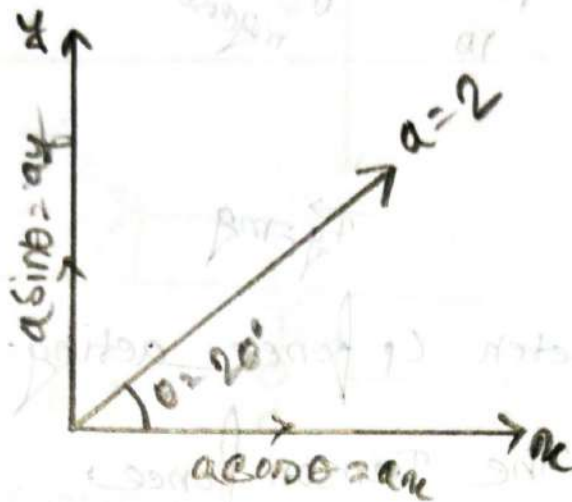
So that in this way we can Explain that Newton 1st law from 2nd law.

Q) is given,

(a)

$$a = 2 \text{ ms}^{-2} \quad \left\{ \begin{array}{l} m = 1 \text{ kg} \\ \theta = 20^\circ \end{array} \right.$$

we know,



$$\begin{aligned} a_x &= a \cos \theta \\ &= 2 \times \cos 20^\circ \\ &= 1.88 \text{ ms}^{-2} \end{aligned}$$

So, $F_x = m a_x$

$$\begin{aligned} &= 1 \times 1.88 \\ &= 1.88 \text{ N} \end{aligned}$$

we know,

(b)

$$\begin{aligned} a_y &= a \sin \theta \\ &= 2 \times \sin 20^\circ \\ &= 0.69 \text{ ms}^{-2} \end{aligned}$$

So $F_y = m a_y$

$$\begin{aligned} &= 1 \times 0.69 \\ &= 0.69 \text{ N} \end{aligned}$$

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The net force in Unit-Vector notation, is

$$\begin{aligned}\vec{F} &= F_x \hat{i} + F_y \hat{j} \\ &= 1.88 \hat{i} + 0.69 \hat{j}\end{aligned}$$

Ans:

Problem 833

→ An elevator cab and its load have a combined mass of 1600 kg. Find the tension in the supporting cable when the cab, originally moving downwards at 12 m/s, is brought to rest with constant acceleration in a distance of 42 m.

→ we have,

$$m = 1600 \text{ kg}$$

$$v = 12 \text{ m/s}$$

$$d = 42 \text{ m}$$

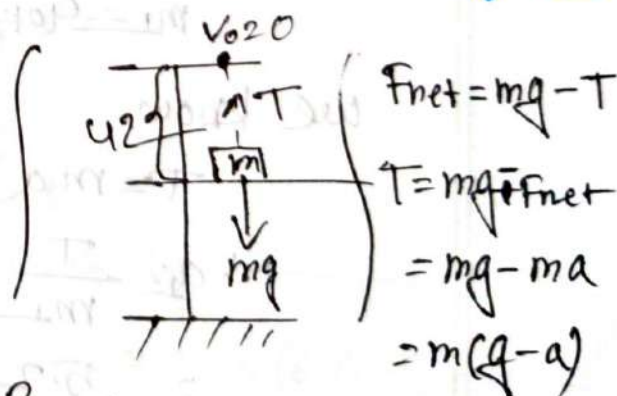
we know,

$$T = mg - ma$$

$$= m(g - a)$$

$$= 1600(9.8 - 1.714)$$

$$= 12937 \text{ N}$$



$$\begin{aligned}v^2 &= u^2 + 2as \\ a &= \frac{v^2 - u^2}{2s} \\ &= \frac{(12)^2 - 0}{2 \times 42} \\ &= 1.714 \text{ m/s}^2\end{aligned}$$

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Problem: 378

⇒ A 40 kg girl and an 8.4 kg sled are on the frictionless ice of a frozen lake, 15 m apart but connected by a rope of negligible mass. The girl exerts a horizontal 5.2 N force on the rope.

- (a) What are the acceleration magnitudes of the sled and girl.
(b) How far from the girl's initial position do they meet.

For girl:

we have,

$$T = 5.2 \text{ N}$$

$$m_1 = 40 \text{ kg}$$

we know,

$$T = m a$$

$$\therefore a_1 = \frac{T}{m_1}$$

$$= \frac{5.2}{40}$$

$$= 0.13 \text{ m/s}^2$$

(a)

For sled:

we have,

$$T = 5.2 \text{ N}$$

$$m_2 = 8.4 \text{ kg}$$

we know,

$$T = m_2 a_2$$

$$\therefore a_2 = \frac{T}{m_2}$$

$$= \frac{5.2}{8.4}$$

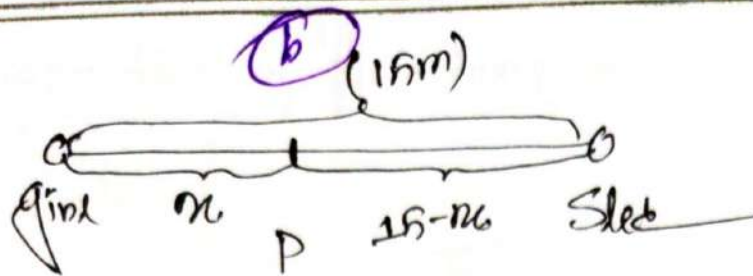
$$= 0.62 \text{ m/s}^2$$

Ans

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We assume that they will meet at point c after a time t .

we have,

$$a_1 = 0.13 \text{ ms}^{-2}$$

$$a_2 = 0.610 \text{ ms}^{-2}$$

for girl —
from

$$S = ut + \frac{1}{2} a_1 t^2$$

$$x = ut + \frac{1}{2} a_1 t^2$$

$$x = 0 + \frac{1}{2} a_1 t^2$$

$$x = \frac{1}{2} a_1 t^2 \quad \text{--- (1)}$$

for sled —
from

$$S = ut + \frac{1}{2} a_2 t^2$$

$$15 - x = 0 + \frac{1}{2} a_2 t^2$$

$$15 - \frac{1}{2} a_1 t^2 = \frac{1}{2} a_2 t^2$$

$$\frac{30 - a_1 t^2}{2} = \frac{1}{2} a_2 t^2$$

$$30 - a_1 t^2 = a_2 t^2$$

$$t = \sqrt{\frac{30}{a_1 + a_2}}$$

$$= 6.320 \text{ Sec}$$

So,

$$x = \frac{1}{2} a_1 t^2$$

$$= \frac{1}{2} \times 0.13 \times (6.320)^2$$

$$= 2.604 \text{ m}$$

Ans

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Chapter 06: Motion

Force

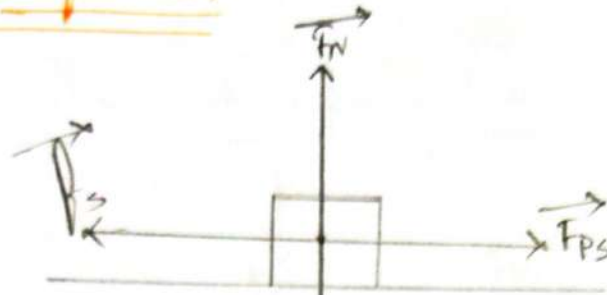
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Properties of friction:

1. Weak Applied force,

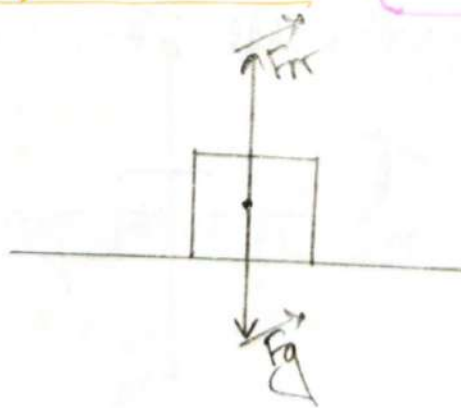


$F_{ps} = F_s$

$$\text{if } F_{ps} = F_s$$

box remains at rest.

2. No Applied force

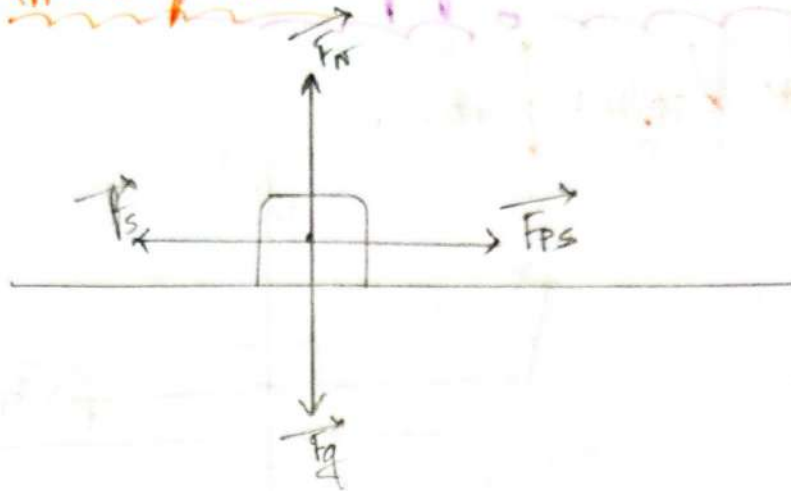


$$\text{if } F_{ps} = 0$$

No, friction, $F_s = 0$

box at rest.

Stronger applied force, start to accelerate the

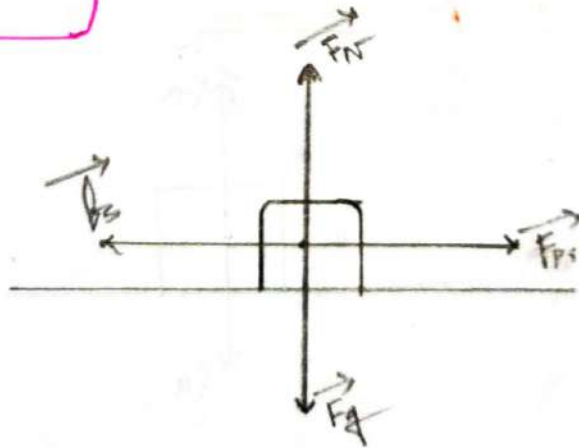


$$f_{s, \text{max}} = \mu_s F_N$$

Box just about to slide.

Box sliding at approximately constant speed:

$$f_k = \mu_k F_N$$

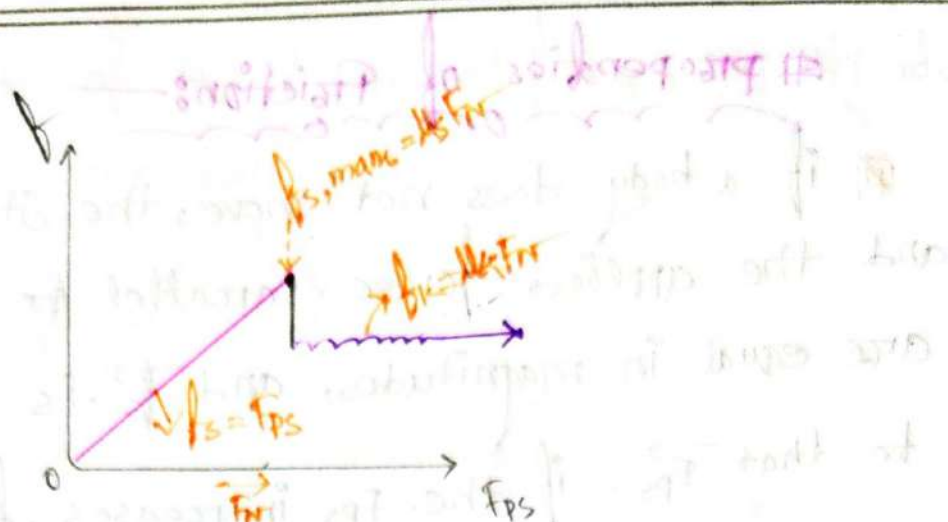


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Friction

- ⇒ When a force tends to slide a body along a surface, a frictional force from the surface acts on the body. The frictional force is parallel to the surface (F_{ps}) and directed so as to oppose the sliding; it is due to bonding between the body and surface.
- ⇒ if the body does not slide, the frictional force is a **Static frictional force (f_s)**.
- ⇒ if there is sliding, the frictional force is a **Kinetic frictional force (f_k)**.

Properties of Friction:

☐ If a body does not move, the static frictional force ($\vec{f}_s$) and the applied force parallel to the surface ($\vec{F}_s$) are equal in magnitude, and $\vec{f}_s$ is directed opposite to that $\vec{F}_s$. If the F_s increases, f_s also increases.

☐ The magnitude of $\vec{f}_s$ has a maximum value $f_{s, \text{max}}$ that is given by

$$f_{s, \text{max}} = \mu_s F_N$$

Where μ_s is the Coefficient of static friction and F_N is the magnitude of the Normal force. On the body from the surface. If the magnitude of the $\vec{F}_s$ exceeds $f_{s, \text{max}}$, then the body begins to slide along the surface.

μ_s = coefficient of static friction.

F_N = magnitude of the normal force.

☐ If the body begins to slide along the surface,

+ magnitude of the frictional force rapidly decreases to a value f_k given by

$$f_k = \mu_k f_N$$

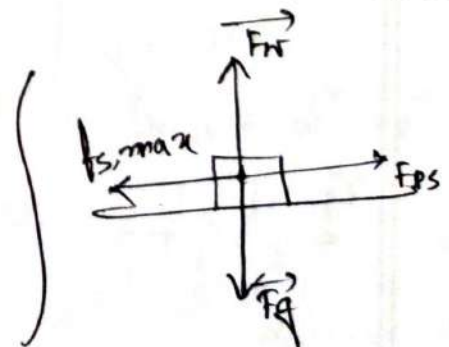
where, μ_k = Coefficient of kinetic friction.
 f_N = Magnitude of the normal force.

Problem 1

The floors of a railroad flatcar is loaded with loose crates having a coefficient of static friction of 0.25 with the floors. If the train is initially moving at a speed of 48 km/h, in how short a distance can the train be stopped at constant deceleration without causing the crates to slide over the floors.

⇒ Since the crates are not sliding, the net force on the crate along horizontal axis (x-axis) is zero. So that, we can write,

$$\begin{aligned} \vec{F}_{\text{net}} &= \vec{F}_{ps} - \vec{f}_{s, \text{max}} \\ 0 &= \vec{F}_{ps} - \vec{f}_{s, \text{max}} \\ \Rightarrow \vec{F}_{ps} &= \vec{f}_{s, \text{max}} \end{aligned}$$



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$$ma = \mu_s \times F_N$$

$$ma = \mu_s \times mg$$

$$a = \mu_s \times g$$

$$= 0.25 \times 9.8$$

$$= 2.45 \text{ m/s}^2$$

We know,

$$v^2 = v_0^2 - 2as$$

$$0 = (13.33)^2 - 2 \times 2.45 \times s$$

$$s = \frac{0 - (13.33)^2}{-2 \times 2.45}$$

$$= 36.26 \text{ m}$$

Ans:-

$$* f_s = \mu_s F_N = \mu_s F_g = \mu_s W = \mu_s mg$$

$$* f_{ok} = \mu_k F_N = \mu_k F_g = \mu_k W = \mu_k mg$$

$$F_{ps} = ma$$

$$f_{s, \max} = \mu_s \times F_N$$

$$= \mu_s \times mg$$

$$F_N = mg$$

$$\mu_s = 0.25$$

$$g = 9.8$$

$$v = 0$$

$$v_0 = 48 \times 10^3 / 3600$$

$$= 13.33 \text{ m/s}$$



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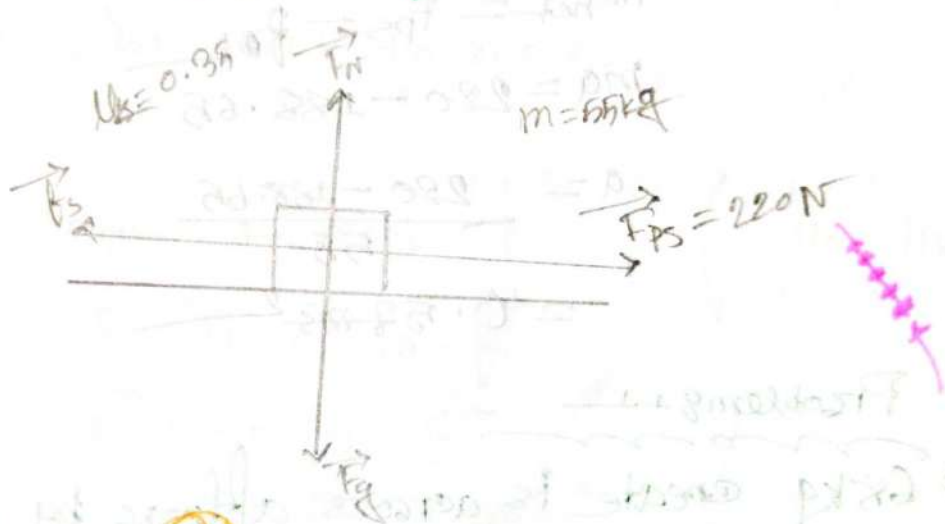
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Problems 07%

A person pushes horizontally with a force of 220 N on a 55 kg crate to move it across a level floor. The coefficient of kinetic friction between the crate and the floor is 0.35 .

- What is the magnitude of the frictional force?
- What is the magnitude of the acceleration of the crate?



We know,

$$f_k = \mu_k F_N = \mu_k mg$$

$$= 0.35 \times 55 \times 9.8$$

$$= 188.65\text{ N}$$

$$\left. \begin{array}{l} \mu_k = 0.35 \\ m = 55\text{ kg} \\ g = 9.8 \end{array} \right\}$$

$$F_{\text{net}} = F_N - F_g$$

$$0 = F_N - F_g$$

$$F_N = F_g$$

$$= mg$$

(b)

we have,

$$F_{ps} = 220 \text{ N}$$

$$f_k = 188.65 \text{ N}$$

$$m = 55 \text{ kg}$$

$$\mu_k = 0.35$$

$$g = 9.8 \text{ m/s}^2$$

we know,

$$\vec{F}_{\text{net}} = \vec{F}_{ps} - \vec{f}_k$$

$$ma = 220 - 188.65$$

$$a = \frac{220 - 188.65}{55}$$

$$= 0.57 \text{ m/s}^2$$

Problems

⇒ 68 kg crate is across a floor by pulling on a rope attached to the crate and inclined 25° above the horizontal.

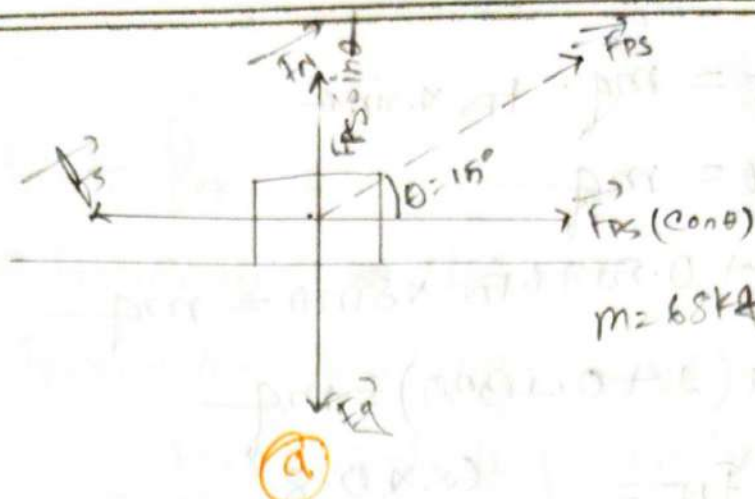
(a) If the coefficient of static friction is 0.50, what minimum force magnitude is required from the rope to start the crate moving?

(b) if $\mu_k = 0.35$ what is the magnitude of the initial acceleration of the crate?

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we can write

The crate facing maximum static frictional force ($f_{s, \text{max}}$) because it is just start to move.

we know,

$$\begin{aligned} F_{\text{net}} &= F_{\text{ps}} \cos \theta - f_s \\ 0 &= F_{\text{ps}} \cos \theta - f_s \\ F_{\text{ps}} \cos \theta &= \mu_s \times F_N \\ &= 0.50 \times F_N \end{aligned} \quad \left(\mu_s = \mu_k = 0.50 \right)$$

$$\begin{aligned} F_{\text{ps}} &= \frac{0.50 \times F_N}{\cos 15^\circ} \\ &= 0.517 F_N \quad \text{--- (1)} \end{aligned}$$

Again

$$\begin{aligned} F_N + F_{\text{ps}} \sin \theta - F_g &= F_{y, \text{net}} \\ F_N + F_{\text{ps}} \sin \theta &= F_g \end{aligned}$$

$$\begin{aligned} \text{So, } F_N &= F_g - F_{\text{ps}} \sin \theta \\ &= mg - 0.517 F_N \sin \theta \end{aligned}$$

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$$F_N = mg - F_{ps} \times \sin \theta$$

$$F_N + F_{ps} \sin \theta = mg$$

$$F_N + 0.5176 F_N \times \sin \theta = mg$$

$$F_N (1 + 0.1339) = mg$$

$$F_N = \frac{68 \times 0.8}{0.1339 + 1}$$

$$= 587.71 \text{ N} \quad \text{--- (1)}$$

By substituting the value of F_N from equation (1) in equation (1), we get.

$$F_{ps} = 0.517 \times F_N$$

$$= 0.517 \times 587.71$$

$$= 303.84 \text{ N}$$

(b)

Now, we can assume that the crate is moving with an acceleration a .

we have,

$$\mu_k = 0.35$$

$$m = 68 \text{ kg}$$

$$\theta = 15^\circ$$

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We know:-

$$F_{\text{net}} = F_{\text{PS}} - \mu_k$$

$$ma = F_{\text{PS}} \cos \theta - \mu_k F_N$$

$$= F_{\text{PS}} \cos \theta - \mu_k (mg - F_{\text{PS}} \sin \theta)$$

$$a = \frac{F_{\text{PS}} \cos \theta - \mu_k (mg - F_{\text{PS}} \sin \theta)}{m} \quad \text{--- (1)}$$

Now

$$= \frac{303.84 \cos 15^\circ - 0.35 (68 \times 9.8 - 303.84 \sin 15^\circ)}{68}$$

$$= \frac{223.48 - 0.35 \times 587.76}{68}$$

$$= \frac{87.764}{68}$$

$$= 1.29 \text{ m/s}^2$$

Ans:-

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No. of Kinetic Energy and Work

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Kinetic Energy

⇒ Kinetic energy = K

is a energy associated with the state of motion of an object. The faster the objects moves, the greater is its kinetic energy. When the object is stationary, its kinetic energy is zero.

$$K = \frac{1}{2}mv^2$$

Where,

K = Kinetic Energy.

m = mass of the object

v = the speed of light.

S.I units = J (Joule)

$$1 \text{ Joule} = 1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$$

Work

⇒ Work = W is a energy transferred to or from an object via a force acting on the object

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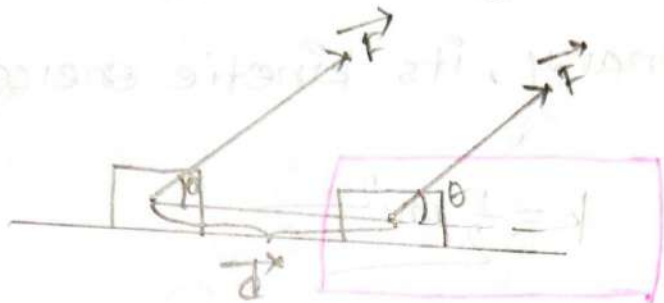
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1. Energy transferred to the object is positive work.

2. Energy transferred from the object is negative work.

if. The work done on a particle by a constant force $\vec{F}$ during displacement $\vec{s}$ is



Work: $W = \vec{F} \cdot \vec{s} = F \times s \cos \theta$

where,

F = Force

s = displacement

θ = Angle.

Condition of positive and negative work

1. Positive work

$\therefore 0^\circ \leq \theta < 90^\circ$

2. Negative work

$\therefore 90^\circ < \theta \leq 180^\circ$

2. No work:

$$\hat{i} \cdot \theta = 90^\circ$$

The principle of work-kinetic energy theorem

$$\Delta K = K_f - K_i = W$$

Where:

ΔK = A change in the kinetic energy.

K_f = final kinetic energy.

K_i = initial kinetic energy.

W = work.

S.I units = J (Joule)

$$1 \text{ Joule} = 1 \text{ N} \cdot \text{m} = 0.738 \text{ ft} \cdot \text{lb}$$

Theorem:

We know, $\Delta K = K_f - K_i = W$

where

$$K_i = \frac{1}{2} m v_i^2$$

$$K_f = \frac{1}{2} m v_f^2$$

Now,

$$W = \Delta K = K_f - K_i$$

$$= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W = \frac{1}{2} m (v_f^2 - v_i^2)$$

Finding an Expression for work-kinetic energy

Deduction

⇒ Using the equation for motion with constant acceleration, we can write.

$$v^2 = v_0^2 + 2axd$$

$$\Rightarrow v^2 - v_0^2 = 2axd$$

$$\Rightarrow a_k = \frac{v^2 - v_0^2}{2d} \quad \text{--- (i)}$$

we know, $F_k = ma_k$ --- (ii)

Substituting the value of a_k in equation (ii)

$$F_k = ma_k$$

$$= m \left(\frac{v^2 - v_0^2}{2d} \right)$$

$$\Rightarrow F_k \times d = \frac{1}{2} mv^2 - \frac{1}{2} mv_0^2$$

$$\Rightarrow F \cos \theta \times d = \frac{1}{2} m(v^2 - v_0^2)$$

$$\Rightarrow Fd \cos \theta = \frac{1}{2} m(v^2 - v_0^2)$$

$$\Rightarrow \boxed{W = \frac{1}{2} m(v^2 - v_0^2)}$$

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$$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$W = K_f - K_i$$

where, $W = F \cdot d \cdot \cos \theta$

$$K_f = \frac{1}{2}mv_f^2$$

$$K_i = \frac{1}{2}mv_i^2$$

Work done by the gravitational forces

We know,

$$W = F \cdot d \cdot \cos \theta$$

Now,

$$W_g = F_g \cdot d \cdot \cos \theta$$

W_g : work done by gravitational force

F_g : gravitation force

d : displacement.

θ : Angle between $\vec{F}_g$ and $\vec{d}$

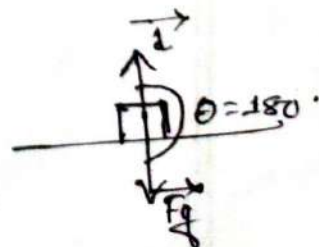
Force raising an object

$\Rightarrow$ Force $\vec{F}_g$ is directed opposite the displacement $\vec{d}$

$$W_g = F_g d \cos \theta$$

$$= F_g d \cos 180^\circ$$

$$W_g = -F_g d$$



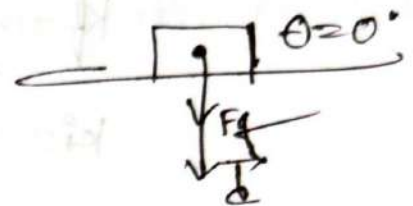
For lowering objects:

$\Rightarrow$ Force $\vec{F}_g$ and displacement $\vec{d}$ is moving same direction.

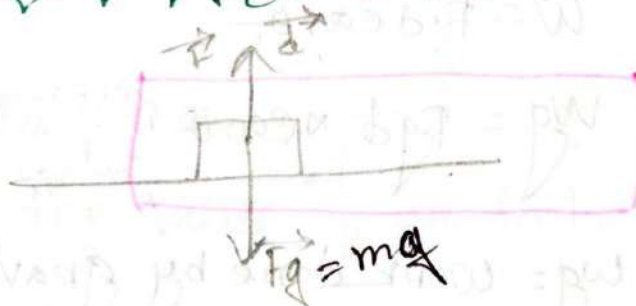
$$\text{So, } W_g = \vec{F}_g \cdot \vec{d} \cos \theta$$

$$= F_g d \cos 0^\circ$$

$$W_g = F_g d$$



Calculating total work:



$$W_1 = F_s \cos \theta$$

$$= F_s \cos 0^\circ$$

$$W_1 = F_s$$

$$W_1 = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$W_2 = F_g \cos 180^\circ$$

$$= -mgs$$

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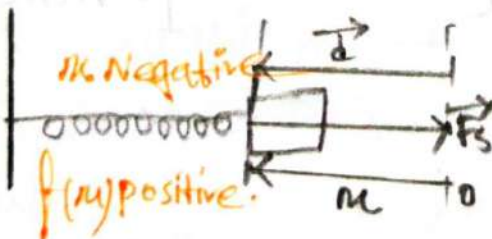
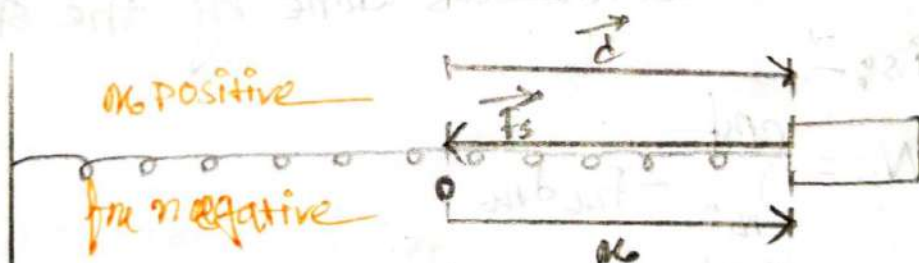
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So total work $W = W_1 + W_2$

$$= \frac{1}{2} m (v_f^2 - v_i^2) + (mgs)$$

$$W = K_E - K_P$$

The spring forces



$$\vec{F}_s = -k\vec{x}$$

k = spring constant

$x = \vec{x}$ = displacement

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The SI unit for k is Nm^{-1}

* if an x -axis lies along the spring, with the origin at the location of the spring's free end when the spring is in its relaxed state, we can write,

$$F_x = -kx$$

The work done by spring forces—

⇒ The net work W_s done by the spring (from x_i to x_f) is:-

$$W_s = \int_{x_i}^{x_f} -F_x dx$$

$$= \int_{x_i}^{x_f} -kx dx$$

[where $|F_x| = kx$]

$$= -k \int_{x_i}^{x_f} x dx$$

$$= -k \left[\frac{x^2}{2} \right]_{x_i}^{x_f}$$

$$= -k \times \frac{1}{2} (x_f^2 - x_i^2)$$

$$W_s = -\frac{1}{2} k (x_f^2 - x_i^2)$$

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if $m_i = 0$ and if $m_f = m$ then,

$$W_s = - \frac{1}{2} k m^2$$

Problems

A proton (mass $m = 1.67 \times 10^{-27} \text{ kg}$) is being accelerated along a straight line at $3.6 \times 10^{15} \text{ m s}^{-2}$ in machine. if the proton has an initial speed of $2.4 \times 10^7 \text{ m/s}$ and travels 3.5 cm ,

(a) What then its speed?

(b) What then is the increase in its kinetic energy?

we have,

$$m = 1.67 \times 10^{-27} \text{ kg}$$

$$a = 3.6 \times 10^{15} \text{ m s}^{-2}$$

$$v_i = 2.4 \times 10^7 \text{ m s}^{-1}$$

$$s = 3.5 \times 10^{-2} \text{ m}$$

we know,

$$v_f^2 = v_i^2 + 2as$$

$$v_f^2 = (2.4 \times 10^7)^2 + 2 \times 3.6 \times 10^{15} \times 3.5 \times 10^{-2}$$

$$v_f^2 = 8.28 \times 10^{14}$$

$$v_f = \sqrt{8.28 \times 10^{14}} = 2.87 \times 10^7 \text{ m s}^{-1}$$

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We know,

The change (increase) in kinetic energy: —

$$W_s = \Delta K = K_f - K_i = \frac{1}{2} m (v_f^2 - v_i^2)$$

$$\Delta K = \frac{1}{2} (1.67 \times 10^{-27}) \left\{ (2.88 \times 10^7)^2 - (2.4 \times 10^7)^2 \right\}$$
$$= 2.11 \times 10^{-13} \text{ J}$$

Ans: —

Problem 9: —

→ The only force acting on a 2.0 kg canister that is moving in an $x-y$ plane has a magnitude of 5.0 N. The canister initially has a ^{velocity} 4.0 m/s in the positive x -direction. and some times later has a velocity of $(6.0) \text{ m/s}$ in the positive y direction.

(a) How much work done on the canister by the 5.0 N force during the time.

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⇒ we have, $m = 2.0 \text{ kg}$
 $v_i = 4 \text{ m/s}$
 $v_f = 6 \text{ m/s}$

we know

$$\begin{aligned} W &= \Delta K = K_f - K_i \\ &= \frac{1}{2} m (v_f^2 - v_i^2) \\ &= \frac{1}{2} \times 2 (6^2 - 4^2) \\ &= 20 \text{ J} \end{aligned}$$

Ans: —

Problem 9.119

A 12.0 N force with a fixed orientation does work on a particle as the particle moves through the three dimensional displacement $\vec{d} = (2\hat{i} - 4\hat{j} + 3\hat{k}) \text{ m}$. What is the angle between the force and the displacement if the change in the particle's kinetic energy is (a) $+30.0 \text{ J}$ and (b) -30.0 J .

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we have

$$F = 12\text{ N}$$

$$W = 30\text{ J}$$

$$\vec{d} = (2\hat{i} - 4\hat{j} + 3\hat{k})$$

$$|\vec{d}| = \sqrt{2^2 + 4^2 + 3^2}$$
$$= \sqrt{29}$$

we know,

$$W = Fd \cos \theta$$

$$\cos \theta = \frac{W}{Fd}$$

$$\theta = \cos^{-1} \left(\frac{W}{Fd} \right)$$

$$= \cos^{-1} \left(\frac{30}{12 \times \sqrt{29}} \right)$$

$$\theta_1 = 62.93^\circ$$

(b)

$$\theta_2 = \cos^{-1} \left(\frac{-30}{12 \times \sqrt{29}} \right)$$
$$= 117.66^\circ$$

$$\left. \begin{array}{l} \text{Given,} \\ W = -30\text{ J} \\ F = 12\text{ N} \\ d = \sqrt{29} \end{array} \right\}$$

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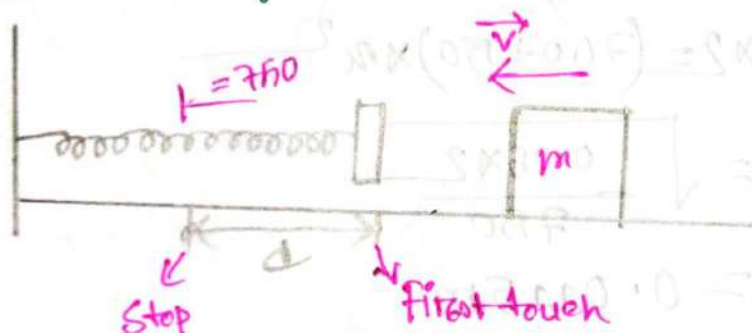
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Problem : 7.06 (Page 161) :

A Cumin Canister of mass $m = 0.40 \text{ kg}$ slides across a horizontal frictionless counter with speed $v = 0.50 \text{ m/s}$. It then runs into and compresses a spring of spring constant $k = 750 \text{ N/m}$. When the canister is momentarily stopped by the spring, by what distance d is the spring compressed?



we have,

$$v = 0.50 \text{ m/s}$$

$$m = 0.40 \text{ kg}$$

$$k = 750 \text{ N/m}$$

we know,
m o m

The Kinetic energy

$$K_e = \frac{1}{2} m v^2$$

$$= \frac{1}{2} (0.40) \times (0.50)^2$$

$$= 0.05 \text{ J}$$

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When the canister is stopped by the spring, All of its kinetic energy is converted into potential energy stored in the spring.

Potential energy = work done by spring

$$P_E = W_s = 0.05$$

we know,
work done

$$W_s = \frac{1}{2} kx^2$$

$$0.05 \times 2 = (700 + 50) x^2$$

$$\Rightarrow x = \sqrt{\frac{0.05 \times 2}{750}}$$

$$= 0.0115 \text{ m}$$

Ans:-

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Ch 8 Potential Energy & Conservation of Energy.

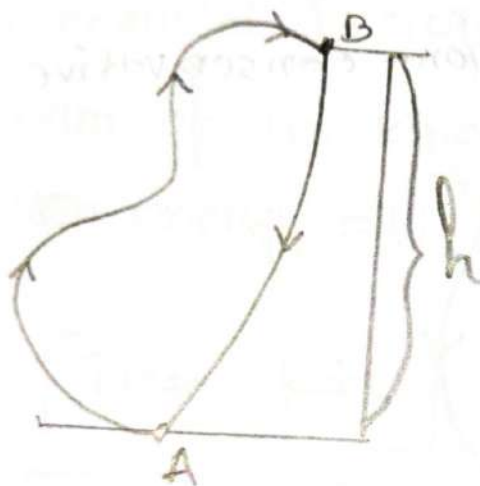
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Conservative forces

⇒ A force is a Conservative force, if the net work it does on a particle moving around any closed path, from an initial point and then back to the point, is zero.



$$A \rightarrow B ; W_1 = -mgh$$

$$B \rightarrow A ; W_2 = mgh$$

$$\begin{aligned} \text{So, Net work } W &= W_1 + W_2 \\ &= -mgh + mgh \\ &= 0 \end{aligned}$$

So, it's a Conservative force.

Non Conservative force :-

⇒ A force that is not conservative is called a non-conservative force.

1. kinetic frictional force. &

2. drag force

Those are non-conservative force.



$$A \rightarrow B ; W_1 = \int \vec{F}_1 \cdot d\vec{s}$$

$$B \rightarrow A ; W_2 = \int \vec{F}_2 \cdot d\vec{s}$$

So, Net work $W = W_1 + W_2$

$$= - \int \vec{F}_1 \cdot d\vec{s} - \int \vec{F}_2 \cdot d\vec{s}$$

$$\neq 0$$

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The principle of Conservation of Mechanical Energy:—

Mechanical Energy:—

⇒ The mechanical Energy = E_{mec} of a system is the sum of its kinetic energy = K and Potential energy = U of the objects within it.

So,

$$E_{mec} = K + U$$

$$\Delta E_{mec} = \Delta K + \Delta U = 0$$

When $\Delta K = +W$

$$\Delta U = -W$$

$$\Delta E_{mec} = E_K + E_P$$

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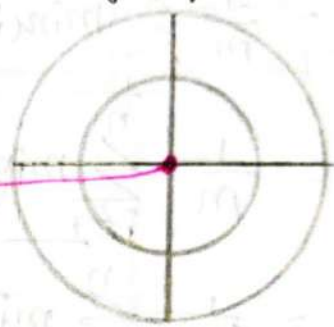
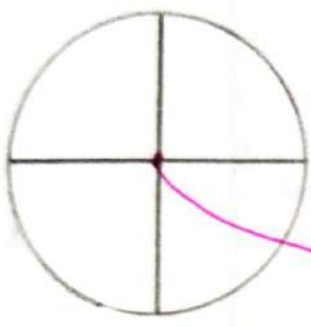
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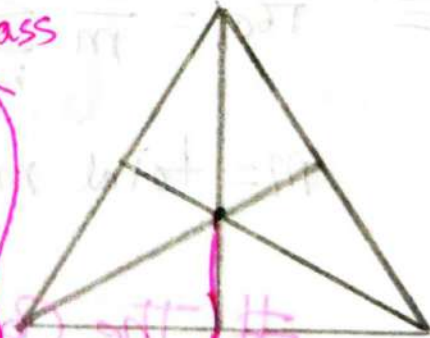
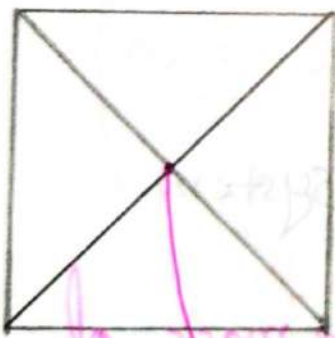
Choo Centers of Mass and Linear Momentum

#Center of Mass

⇒ The center of mass is the average position of the mass of an object or system, and is used to analyze the motion and stability of the system.



The Centers
of
the mass



How to find the Center of mass:-

⇒ The Center of mass of a system of n particles is defined to be the point whose coordinates are given by

$$x_c = \frac{1}{m} \sum_{i=1}^n m_i x_i$$

$$y_c = \frac{1}{m} \sum_{i=1}^n m_i y_i$$

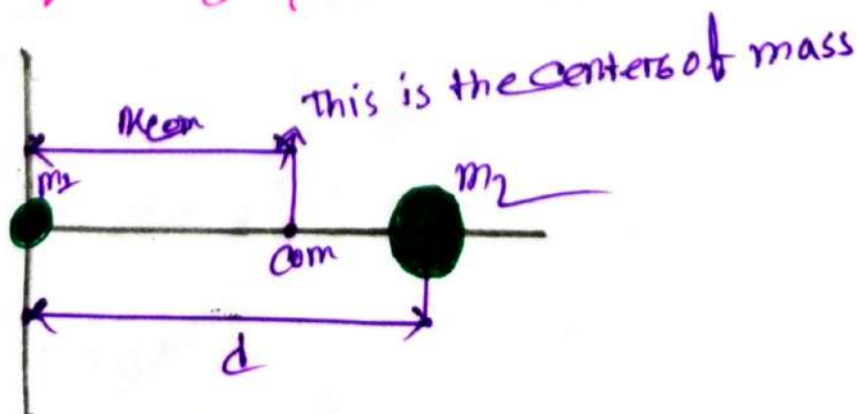
$$z_c = \frac{1}{m} \sum_{i=1}^n m_i z_i$$

So, $\vec{r}_c = \frac{1}{m} \sum_{i=1}^n m_i \vec{r}_i$

m = total mass of the system.

The Center of the mass of the two-particle system:-

Case 1:-



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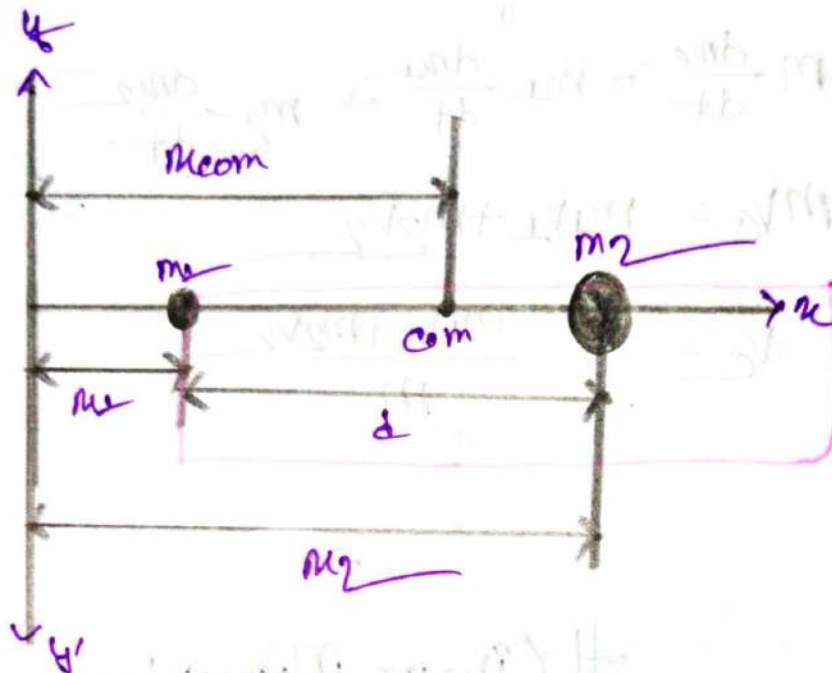
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$$x_c = \frac{m_2 d}{m_1 + m_2}$$

Case 2:



$$x_c = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

& The Velocity of the Center of mass

we have,

Center of mass $x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$

$$= \frac{m_1 x_1 + m_2 x_2}{m}$$

$M = m_1 + m_2$

$$m v_c = m_1 v_1 + m_2 v_2$$

Differentiating with respect to time gives

$$m \frac{dv_c}{dt} = m_1 \frac{dv_1}{dt} + m_2 \frac{dv_2}{dt}$$

$$m v_c = m_1 v_1 + m_2 v_2$$

$$v_c = \frac{m_1 v_1 + m_2 v_2}{m}$$

Linear Momentum

⇒ The linear momentum of a particle is a vector quantity $\vec{p}$ that is defined as

$$\vec{p} = m \vec{v}$$

m = mass of the particle

v = velocity of the particle.

A particle's momentum $\vec{p}$ has the same direction as its velocity $\vec{v}$.

SI Units = kg m s^{-1}

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Force and Momentum

$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt}$$

$$\vec{F} = m\vec{a}$$

which is the Newton's second law in terms of momentum.

The Linear Momentum of a System of Particles:

$$\vec{P} = m\vec{v}_c$$

$\vec{P}$ = The linear momentum.

m = the total mass

$\vec{v}_c$ = the Velocity of the Center of mass.

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Collision and Impulse

We know,

$$\vec{F} = \frac{d\vec{p}}{dt}$$

in time interval dt , the change in the ball's momentum is

$$d\vec{p} = \vec{F}(t) dt$$

SI unit of = N s

magnitude of = MLT^{-1}

Relation of impulse to momentum:-

$$\vec{J} = \int_{t_i}^{t_f} \vec{F} dt$$

$$\vec{J} = \int_{\vec{p}_i}^{\vec{p}_f} \frac{d\vec{p}}{dt} \times dt$$

$$\left[\begin{array}{l} \text{We know,} \\ \vec{F} = \frac{d\vec{p}}{dt} \end{array} \right]$$

$$\vec{J} = \left[\vec{p} \right]_{\vec{p}_i}^{\vec{p}_f}$$

$$\vec{J} = \vec{p}_f - \vec{p}_i$$

$$\vec{J} = \Delta \vec{p}$$

Collision:

If F_{avg} = The Average Magnitude of $\vec{F}(t)$ during the collision.

Δt = duration of the collision.

$$\vec{J} = F_{avg} \Delta t$$

The law of Conservation of linear momentum

⇒ If a system is closed and isolated so that no net external force ^{acts} on it, then the linear momentum must be constant even if there are internal change.

$$\vec{P} = \text{constant}$$

That means $\vec{P}_i = \vec{P}_f$

In words, this equation says that, for a closed, isolated system.

$$(\text{Total linear momentum at some initial time } t_i) = (\text{Total linear momentum at some later time } t_f)$$

Short Q:- What is the difference between momentum and linear momentum?

Momentum—

→ Think of momentum as a measure of how hard it is to stop something that's moving. An object with more momentum will be harder to stop than an object with less momentum.

Momentum takes into account an object's mass and how fast it's moving.

Linear Momentum!—

→ It's just a type of momentum that only looks at an object's motion in a straight line. It's like measuring how hard it is to stop something that's moving in a straight line. So if an object is moving in a straight line, it has linear momentum, but if it's moving in a curve, it doesn't have linear momentum.

In words, momentum is like a measure of how hard it is to stop something that's moving,

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While linear momentum is just a measure of how hard it is to stop something that moving in a straight line.
